

App Dev Project Report

1. Student Details

Name: Neha Varshney

Roll Number: 24f3003158

Email: 24f3003158@ds.study.iitm.ac.in

About me: I am BS degree(data science) student in IITM. I have deep interest in learning web application development and building web apps.

2. Project Details

Project Title: Trekking Management Portal

Problem Statement:

To design for book slots by the users, and it helps a lot of admin for assign staffs, add new trek and add new staff etc.

Approach:

The app was built using Flask as the backend framework with a modular structure and [vue.js](#) for UI. It allows users to book their slots to the best available treks and desired treks.

3. AI/LLM Declaration

I used ChatGPT (GPT-5) to assist in writing SQLAlchemy model definitions and for creating backend. The extent of AI/LLM usage is around 15-20%, limited to code suggestions and documentation formatting.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library Purpose

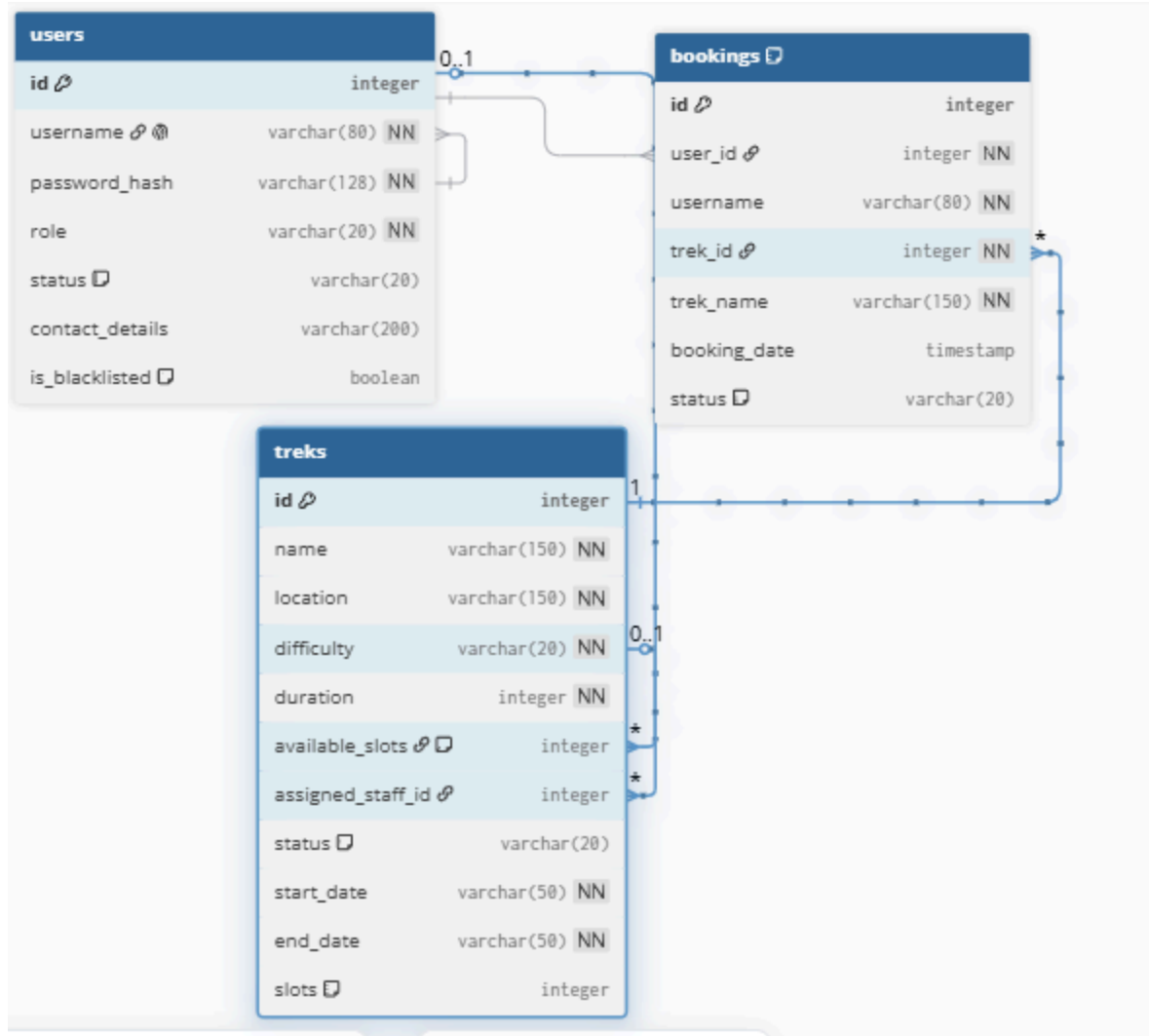
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Flask-Login	User authentication and session management
WTForms	Frontend form validation
SQLite	Lightweight local database for storing user data
Redis	Caching
Vue.js	for UI

5. Database Schema / ER Diagram

Tables:

1. User — stores user profile details (id, username,status, password_hash,role,contact_details)
2. Trek —it contains trekking details (id,name,location,difficulty ,duration, available_slots,assign_staff_id,status,start_date,end_date, slots)
- 3.Booking —it contains booking details
(id,user_id,trek_id,status,username,trek_name,booking_date,status)

Relationships:



6. API Resource Endpoints

Endpoint	Method	Description
/api/login	POST	Authenticate user and generate session
/api/activities	GET	Fetch logged activities for a user
/api/activity/add	POST	Add new activity entry

/api/summary

GET

Return daily/weekly analytics summary

7. Architectures and features(optional)

Architecture Overview:

- app.py – main Flask application entry point
- /models – database models using SQLAlchemy
- /routes – Flask Blueprints for user and activity routes
- /templates – Jinja2 HTML templates
- /static – CSS and chart visualization files

Implemented Features:

User registration and login
Daily activity tracking and habit logging
Visualization dashboard using Chart.js
Summary analytics with date filtering
Responsive UI built with Bootstrap
Secure password hashing and user session handling

Additional Features:

- Export user logs as CSV
- AI-generated weekly summary insights (optional extension)

8.Video presentation

My video link

<https://drive.google.com/file/d/1hCYbZYSJgbvJCu86fSY3kH5TzUCHO5Sz/view?usp=sharing>